Executive Summary: Lead-Lag Trading

SPY → AAPL and SPY → NVDA | MGMTMFE 412 — Market Frictions

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Intuition

SPY is the most liquid security in the world. When a large buy order hits the SPY book, the price adjusts in *microseconds*. But AAPL and NVDA — even though they are components of the same index — take **1 to 3 seconds longer** to reflect that move.

Why? Because the price transmission mechanism requires human or algorithmic intermediaries to notice the SPY move, calculate its implication for individual stocks, and place orders. That process is fast, but not instantaneous.

This creates a brief, predictable window: **SPY moves first, the constituent stocks follow**. If you can observe SPY moving up and enter AAPL before it catches up, you capture a small but consistent profit.

Strategy

We use Level-2 order book data (Databento MBP-10) at **1-second resolution** for January–April 2024. The signal fires when two conditions hold simultaneously:

1. SPY's 1-second return exceeds **1.0 bps** — a meaningful directional move
2. SPY's order flow imbalance (OFI) exceeds **0.40** — buy-side pressure confirms the move is not noise

When the signal fires, we enter the constituent stock (AAPL or NVDA) in the same direction as SPY and hold for **5 seconds**, then exit. Position size: 100 shares.

Results

The lead-lag signal is **statistically real**. Gross PnL is positive in both stocks in both the training and test periods — the pattern is not random.

But here is the central finding: **transaction costs destroy all the alpha**.

To enter and exit a trade, we must cross the bid-ask spread twice — once on the way in, once on the way out. For AAPL this costs about **$1.18 per trade** on average. The gross profit per trade is only **$0.13**. Transaction costs are **9 times larger** than the gross edge.

	AAPL	NVDA
Gross PnL (train + test)	+$42	+$1,038
Transaction costs	−$376	−$6,679
Net PnL	**−$334**	**−$5,641**
TC as multiple of gross alpha	**9×**	**7.5×**

NVDA has a longer lag (1–3 seconds vs 1 second for AAPL) and generates 20× more gross PnL — but its spread is 4× wider, producing a similarly lopsided TC/PnL ratio. This reveals a fundamental tradeoff: the stocks with the longest exploitable lag always have the highest transaction costs, because *slower price discovery and lower liquidity go hand in hand*.

The Break-Even Insight

For this strategy to be profitable, a trader would need transaction costs below **$0.0013 per share** — roughly 10 times cheaper than what a typical institutional investor pays. This level is only achievable by co-located HFT firms that post limit orders and collect maker rebates rather than crossing the spread.

> **The bid-ask spread is the friction that prevents retail and institutional traders from exploiting a real market inefficiency.** The edge exists — it just belongs entirely to HFT.

Connection to Market Frictions

This result is a textbook illustration of the course's central theme. The lead-lag relationship represents a genuine price inefficiency: information in SPY is not instantly reflected in its components. But the cost of exploiting it — the bid-ask spread — perfectly screens out all but the fastest, cheapest participants. The market is not perfectly efficient; it is *efficiently protected by friction.*